1 Numerical Exercise: Geodesic Integration in Kerr Spacetime

All quantities in geometric units $G = c = 1$, lengths and times in units of M. Derivatives expressed with a dot denote derivative with respect to affine parameter, i.e. $\dot{x}^\mu = \frac{dx^\mu}{d\lambda}$ is the 4-velocity.

1.1 Overview and Objectives

In this exercise you will integrate timelike geodesics in the Kerr spacetime using Kerr–Schild (KS) coordinates and verify the conservation of four integrals of motion. By the end you should be able to:

- implement the KS geodesic ODE and integrate it with a standard solver;
- monitor all four diagnostics (E, L_z, $\mathcal{Q}$, $\mathcal{N}$) to assess numerical accuracy;
- recognise the visual signatures of zoom-whirl orbits, periapsis precession, and orbital-plane precession.

We will proceed with various steps with increasing complexity.

1.2 The ODE System and the Derivative Ladder

The state vector is $\mathbf{Y} = (x, y, z, v^x, v^y, v^z)$ with $v^i = dx^i/dt$. The geodesic equation, rewritten as a 6-dimensional first-order system in coordinate time, reads

$$\frac{d\mathbf{Y}}{dt} = \begin{pmatrix} v^i \\ F^i \end{pmatrix}, \qquad F^i = \bar{F}^i + \frac{f}{2}\,\mathcal{G}\,(l^i + v^i), \tag{1}$$

where $\bar{F}^i$ collects the flat-space and metric-gradient terms, $f = 2Mr^3/(r^4 + a^2z^2)$ is the KS scalar, and $\mathcal{G}$ is a scalar assembled from metric derivatives. All right-hand-side quantities are computed via the *derivative ladder*: starting from the Cartesian KS coordinates (x, y, z) one evaluates, in order,

1. the KS radial coordinate r
2. the null vector l^μ and the scalar f;
3. the metric $g_{\mu\nu} = \eta_{\mu\nu} + f l_\mu l_\nu$ and its spatial gradients $\partial_i g_{\mu\nu}$;
4. the Christoffel symbols $\Gamma^\mu_{\alpha\beta}$ and hence $\bar{F}^i$ and $\mathcal{G}$.

Use `scipy.integrate.solve_ivp` with `method='DOP853'`, `rtol=1e-11`, `atol=1e-13`.

1.3 Integrator test: Schwarzschild Circular Orbit

Implement the ODE system of Section 1.2 for $a = 0$ and integrate the following orbit.

Parameter	Value
M	1
a	0 (Schwarzschild)
e	0.0
p	$50\,M$

Try this initial KS Cartesian state vector (prograde orbit, starting at $\phi = 0$):

$$\mathbf{Y}_0 = \Big(50,\ 0,\ 0,\ 0,\ 0.14142136,\ 0\Big). \tag{2}$$

1. Integrate for at least 5 radial periods. Plot the orbit in the x-y plane and verify that the shape is the correct one.
2. Verify $|\mathcal{N}(0)| \lesssim 10^{-12}$ and plot $|\mathcal{N}(t) - \mathcal{N}(0)|$ as a function of t.
3. Check that E and L_z (in this case also the total angular momentum) are conserved to the level of the integrator tolerance.

1.4 Generic initial Conditions

General procedure

The cleanest strategy is to specify initial data in Boyer–Lindquist (BL) coordinates, where the physics is transparent, and then convert to KS.

Step 1 — choose orbital parameters. Specify the semi-latus rectum p, eccentricity e, inclination. The apocenter and pericenter are

$$r_{\rm apo} = \frac{p}{1-e}, \qquad r_{\rm peri} = \frac{p}{1+e}.$$

Starting at apocenter or pericenter sets $\dot{r} = 0$, which is the recommended choice.

Step 2 — recover E, L_z, $\mathcal{Q}$. For Schwarzschild ($a = 0$), equatorial ($\theta = \pi/2$) bound orbits:

$$E = \sqrt{\frac{(p-2-2e)(p-2+2e)}{p(p-3-e^2)}}, \qquad L_z = \frac{p}{\sqrt{p-3-e^2}}, \qquad \mathcal{Q} = 0 \text{ (equatorial)}. \tag{3}$$

For Kerr the analogous closed-form expressions are given in van de Meent (2019, see appendix A here. In the reference, r_1, r_2 are $r_{\rm apo}, r_{\rm peri}$. You also find the formulae as an appendix of this exercise).

Step 3 — recover the BL 4-velocity. From the first-order BL equations, we get the components of the 4-velocity:

$$\dot{t} = \frac{(r^2+a^2)[E(r^2+a^2) - aL_z]/\Delta - a(aE\sin^2\theta - L_z)}{\Sigma}, \tag{4}$$

$$\dot{\phi} = \frac{a[E(r^2+a^2) - aL_z]/\Delta - (aE - L_z/\sin^2\theta)}{\Sigma}, \tag{5}$$

$$\dot{r} = \frac{\sqrt{R(r)}}{\Sigma}, \tag{6}$$

$$\dot{\theta} = \frac{\sqrt{\Theta(\theta)}}{\Sigma}, \tag{7}$$

where

$$R(r) = [E(r^2+a^2) - aL_z]^2 - \Delta[r^2 + (L_z - aE)^2 + \mathcal{Q}], \tag{8}$$

$$\Theta(\theta) = \mathcal{Q} - \cos^2\theta\big[a^2(1-E^2) + L_z^2/\sin^2\theta\big]\,. \tag{9}$$

and

$$\Delta = r^2 - 2Mr + a^2, \qquad \Sigma = r^2 + a^2\cos^2\theta.$$

At apocenter/pericenter, $\dot{r} = 0$ by definition. The coordinate velocities in BL are $v^\alpha_{\rm BL} = \dot{q}^\alpha/\dot{t}$, where $q^\alpha = (t, r, \theta, \phi)$ are the coordinates in BL.

Step 4 — transform position and velocity to KS. The BL-to-KS position map (a along z-axis) is

$$x = \sqrt{r^2 + a^2}\sin\theta\cos\phi, \quad y = \sqrt{r^2 + a^2}\sin\theta\sin\phi, \quad z = r\cos\theta.$$

The coordinate-velocity transformation is

$$\boxed{v^i_{\mathrm{KS}} = \frac{J^i{}_\alpha\, v^\alpha_{\mathrm{BL}}}{1 + \dfrac{2Mr}{\Delta}\, v^r_{\mathrm{BL}}},} \tag{10}$$

where $J^i{}_\alpha = \partial x^i/\partial q^\alpha$ is the Jacobian of the position map. At apocenter/pericenter $v^r_{\mathrm{BL}} = 0$, so the denominator is exactly 1 and the transformation reduces to a simple Jacobian multiplication.

Step 5 — verify initial data. Before starting the integration check that $|\mathcal{N}(0)| \lesssim 10^{-12}$ (see Section 1.5). A non-zero value indicates an error in the initial data.

1.5 Conservation Laws

The role of $\dot{t}$

The state vector contains only the spatial coordinate velocity $v^i = dx^i/dt$. The 4-velocity is $u^\mu = \dot{t}(1, v^i)$, where $\dot{t} = dt/d\lambda$. Since $\dot{t}$ is *not* evolved by the ODE, it must be recovered at each output step.

The cleanest approach is to use the conserved energy. At $t = 0$ compute

$$\dot{t}_0 = \frac{1}{\sqrt{|g_{\mu\nu}(1, v_0^i)(1, v_0^j)|}}, \tag{11}$$

fix

$$E_0 = -g_{\mu t}u^\mu\big|_{t=0} = \dot{t}_0\big(1 - f(1 + l_i v^i)\big)\big|_{t=0}, \tag{12}$$

and thereafter recover $\dot{t}$ by inverting the energy expression:

$$\boxed{\dot{t}(t) = \frac{E_0}{1 - f\big(1 + l_i v^i\big)}.} \tag{13}$$

Normalization $\mathcal{N}$

$$\mathcal{N} + 1 = g_{\mu\nu}u^\mu u^\nu + 1 = \dot{t}^2\big[-(1-f) + 2f l_i v^i(-1) + (v^i)^2 + f(l_i v^i)^2\big] + 1, \tag{14}$$

where $\dot{t}$ is taken from eq. (13). Monitor the dimensionless drift $|\mathcal{N}(t) - \mathcal{N}(0)|$.

Energy E

Killing vector $k^\mu = (1, 0, 0, 0)$

$$E = -u^\mu k_\mu = -u^\mu g_{\mu\alpha}k^\alpha = \dot{t}\big(1 - f(1 + l_i v^i)\big)\,. \tag{15}$$

Axial angular momentum L_z

Killing vector $R^\mu = (0, 0, y, -x)$

$$L_z = u^\mu R_\mu = u^\mu R^\nu g_{\mu\nu} = \dot{t}\left(x v^y - y v^x - \frac{af(x^2 + y^2)}{r^2 + a^2}(1 + l_i v^i)\right). \tag{16}$$

Carter constant $\mathcal{Q}$

A symmetric tensor $K_{\mu\nu}$ is a Killing tensor if $\nabla_{(\alpha}K_{\mu\nu)} = 0$. In practice the explicit Kerr Killing tensor is obtained not by solving this equation directly, but by demanding separability of the Hamilton–Jacobi equation $g^{\mu\nu}\partial_\mu S\,\partial_\nu S = -m^2$ in BL coordinates, which produces a separation constant identified with $\mathcal{Q} = K_{\mu\nu}u^\mu u^\nu - (L_z - aE)^2$. The Killing tensor reads

$$K_{\mu\nu} = r^2 g_{\mu\nu} + 2\Sigma\, k_{(\mu}n_{\nu)}, \qquad \Sigma = r^2 + a^2\cos^2\theta,$$

where k^μ and n^μ are the principal null directions of Kerr. In KS coordinates the contravariant form is

$$K^{\mu\nu} = \Sigma\big(k^\mu n^\nu + k^\nu n^\mu\big) + r^2 g^{\mu\nu}, \tag{17}$$

with

$$k^\mu = \left(\frac{r^2+2Mr+a^2}{\Delta},\ \frac{xr+ya}{r^2+a^2} + \frac{2ay}{\Delta},\ \frac{yr-xa}{r^2+a^2},\ \frac{2ax}{\Delta}\frac{z}{r}\right), \tag{18}$$

$$n^\mu = \left(\frac{\Delta}{2\Sigma},\ -\frac{\Delta}{2\Sigma}\frac{xr+ya}{r^2+a^2},\ -\frac{\Delta}{2\Sigma}\frac{yr-xa}{r^2+a^2},\ -\frac{\Delta}{2\Sigma}\frac{z}{r}\right), \tag{19}$$

and the inverse metric

$$g^{\mu\nu} = \eta^{\mu\nu} - \frac{2Mr^3}{r^4+a^2z^2}\,l^\mu l^\nu, \qquad l^\mu = \left(-1,\ \frac{rx+ay}{r^2+a^2},\ \frac{ry-ax}{r^2+a^2},\ \frac{z}{r}\right),$$

with $\Delta = r^2 - 2Mr + a^2$.

You can choose: Evaluating $\mathcal{Q} = K_{\mu\nu}u^\mu u^\nu - (L_z - aE)^2$ directly in KS or an equivalent route is to transform the output back to BL at each step and use

$$\mathcal{Q} = p_\theta^2 + \cos^2\theta\left(a^2(1-E^2) + \frac{L_z^2}{\sin^2\theta}\right), \qquad p_\theta = \Sigma\,\dot t\, v^\theta_{\mathrm{BL}}. \tag{20}$$

1.6 Orbit type

Work through the four orbits below in order. All have $M = 1$. Initial conditions are derived from the BL parameters using the procedure of Section 1.4, starting at apocenter with $v^r_{\mathrm{BL}} = 0$.

#	Description	a	p/M	e	ι
1	Schwarzschild eccentric	0	25	0.5	0°
2	Kerr equatorial, mild precession	0.9	10	0.3	0°
3	Kerr equatorial, zoom-whirl	0.9	7	0.7	0°
4	Kerr inclined, orbital precession	0.9	12	0.3	30°

Orbit 1 is the eccentric version of Section 1.3. For orbits 2–4 derive the ICs yourself using eqs. (10) and the van de Meent expressions for E, L_z, $\mathcal{Q}$. Note that inclination ι is defined as $\iota = \pi/2 - \theta$.

> For orbit 4 the inclination $\iota = 30°$ sets the initial $\theta_0 \neq \pi/2$. Start at pericenter with $v^r_{\mathrm{BL}} = 0$; the non-zero v^θ_{BL} is obtained from $\sqrt{\Theta(\theta_0)}/(\Sigma\dot t)$.

For each orbit:

1. plot the trajectory (3-D for orbit 4);
2. plot $|E(t) - E_0|/E_0$, $|L_z(t) - L_{z,0}|/|L_{z,0}|$, $|\mathcal{Q}(t) - \mathcal{Q}_0|/|\mathcal{Q}_0|$, and $|\mathcal{N}(t) - \mathcal{N}(0)|$ on the same axes;
3. comment on which quantity drifts fastest and why.

A Additional formulae

In the following r_1, r_2 represents the turning points of the radial motion, i.e. the solution of

$$\dot{r} = \frac{\sqrt{R(r)}}{\Sigma} = 0. \tag{21}$$

The equation has two additional solution denoted as r_3, r_4. The variable z_1 is instead one of the solution of the polar equation $\dot{z}^2 = 0$ (corresponding to $\theta_{\min}$), recasted in terms of $z = \cos\theta$

$$\dot{z}^2 = \frac{1}{\Sigma}\Big(Q - z^2(a^2(1-E^2)(1-z^2) + L_z^2 + Q)\Big). \tag{22}$$

The constants of motion can the be express with the following expressions

$$E = \sqrt{\frac{\kappa\rho + 2\epsilon\sigma - 2a\sqrt{\frac{\sigma}{a^2}\left(\sigma\epsilon^2 + \rho\epsilon\kappa - \eta\kappa^2\right)}}{\rho^2 + 4\eta\sigma}} \tag{23}$$

$$L_z = -\frac{g(r_2)\mathcal{E} - \sqrt{\left(g(r_2)^2 + h(r_2)f(r_2)\right)\mathcal{E}^2 - h(r_2)d(r_2)}}{h(r_2)} \tag{24}$$

$$Q = z_1^2\left(a^2(1-E^2) + \frac{L_z^2}{1-z_1^2}\right) \tag{25}$$

$$\kappa := d(r_2)h(r_1) - d(r_1)h(r_2) \tag{26}$$

$$\epsilon := d(r_2)g(r_1) - d(r_1)g(r_2) \tag{27}$$

$$\rho := f(r_2)h(r_1) - f(r_1)h(r_2) \tag{28}$$

$$\eta := f(r_2)g(r_1) - f(r_1)g(r_2) \tag{29}$$

$$\sigma := g(r_2)h(r_1) - g(r_1)h(r_2) \tag{30}$$

$$d(r) := \Delta(r)\left(r^2 + a^2 z_1^2\right) \tag{31}$$

$$f(r) := r^4 + a^2\left(r(r+2) + z_1^2\Delta(r)\right) \tag{32}$$

$$g(r) := 2ar \tag{33}$$

$$h(r) := r(r-2) + \frac{z_1^2\,\Delta(r)}{1-z_1^2} \tag{34}$$